Skills Summary

Dilations and Similar Figures

Use this reference while completing your worksheet or when studying.

Skill 1 — Dilating a point about the origin

Rule: a dilation centred at the origin with scale factor k sends $(x, y) \mapsto (kx, ky)$.

Multiply *both* coordinates by the same k. Signs come along unchanged, and $k > 1$ moves the point away from the origin while $0 < k < 1$ moves it closer.

Example: Dilate $(4, -3)$ about the origin with $k = 2$.
Step 1. The centre is the origin, so no subtraction is needed — multiply each coordinate by k.
Step 2. $x' = 2 \cdot 4 = 8$ and $y' = 2 \cdot (-3) = -6 \Rightarrow$ $\mathbf{(8, -6)}$

Try it: Dilate $(-2, 5)$ about the origin with $k = 3$.

check: $\mathbf{(-6, 15)}$

Skill 2 — Finding the scale factor from two lengths

Rule: $k = \dfrac{\text{image length}}{\text{original length}}$

Always divide a length by the length it corresponds to. The same k must work for every pair of matching sides — if two pairs give different numbers, the figures are not similar.

Example: A side of 5 becomes a side of 20. Find k.
Step 1. We know a matching pair of lengths, so divide image by original.
Step 2. $k = \dfrac{20}{5} = 4$, and $4 > 1$ tells us the image is an enlargement. $\Rightarrow$ $\boldsymbol{k = 4}$

Try it: A side of 12 becomes a side of 9. Find k.

check: $\boldsymbol{k = \frac{3}{4}}$

Skill 3 — Why a dilation keeps sides parallel

Rule: $\quad \text{slope} = \dfrac{y_2 - y_1}{x_2 - x_1}$

Dilating multiplies every rise and every run by the same k, so the fraction is unchanged. Equal slopes on two different segments mean the segments are parallel — that is the evidence that a dilation gives a similar figure with matching angles.

Example: $(2, 1)$ to $(4, 4)$, dilated by $k = 2$ to $(4, 2)$ and $(8, 8)$. Compare the slopes.
Step 1. To test for parallel sides, compute the slope of each segment and compare them.
Step 2. Original: $\dfrac{4-1}{4-2} = \dfrac{3}{2}$. Image: $\dfrac{8-2}{8-4} = \dfrac{6}{4}$, which is the same fraction. $\Rightarrow$ both slopes $= \mathbf{\frac{3}{2}}$

Try it: $(1, 3)$ to $(5, 6)$, dilated by $k = 4$ to $(4, 12)$ and $(20, 24)$. Find both slopes.

check: $\mathbf{\frac{3}{4}}$

Skill 4 — Finding a missing length in similar figures

Rule: in similar figures, matching sides are in the same ratio, so

$$\frac{\text{missing image side}}{\text{its original}} = \frac{\text{known image side}}{\text{its original}}$$

Find k from the pair you know, then multiply the other original length by that same k.

Example: A rectangle 5 cm by 8 cm is enlarged so the 5 cm side becomes 15 cm. Find the other side.
Step 1. One matching pair is known, so find k from it and apply it to the side we want.
Step 2. $k = \dfrac{15}{5} = 3$, so the other side is $8 \cdot 3$ cm. $\Rightarrow$ **24** cm

Try it: A rectangle 6 cm by 10 cm is enlarged so the 6 cm side becomes 9 cm. Find the other side, in centimetres.

check: **15**

(!) Watch out: one scale factor must fit every direction. If the width scales by 2 and the height by 3, the picture has been stretched, not dilated — and stretched figures are not similar.